Regime Conditioning Recovers Detection, Not Localisation: An Honest Benchmark for Fault Onset on Load-Varying Fleet Telemetry

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***Abstract*—A change detector on raw haul-truck telemetry spends most of its time detecting the haul cycle: strut pressure jumps when the tray is loaded, brake temperature climbs on every descent, fuel rate tracks the grade. The standard remedy, condition on the operating regime and monitor within-regime residuals, is usually asserted rather than measured. This report measures it, and reports everything the measurement returned, including the parts that argue against the method. A detector-free effect size on NASA C-MAPSS shows the mechanism directly: on six-operating-condition subsets the median fault signature is 0.16 pooled standard deviations but 11.95 within-regime standard deviations (ratios 90.2 and 62.3 on the two subset pairs), and single-condition subsets return exactly 1.00, a negative control nobody designed. At a matched event-counted false-alarm budget, detection falls from 0.93 to 0.17 (and 0.73 to 0.24) when operating conditions go from one to six, and regime-conditional residuals recover 0.95 (and 0.90), above the single-condition references; discovering the regimes by clustering costs nothing over being handed the labels. Three results temper the story. Onset *localisation* is a null: chance-corrected skill differs by -0.08 ± 0.74 paired over five seeds. No false-alarm-reduction claim survives measurement, and it is withdrawn. And on a twelve-rung detector ladder over a physically grounded synthetic fleet, conditioning helps five rungs, leaves five at ceiling, and actively hurts the two boundary-shaped learned novelty detectors, with a regime-label coverage of 0.998 ruling out the obvious explanation. The honest sentence the data supports: regime conditioning helps detectors that measure position or accumulate evidence, and can hurt detectors that score isolation from the healthy cloud. Every number in this report is replayed from committed artifacts, and the engine's own defect record, ten defects that produced plausible numbers rather than crashes, ships as an appendix because it is part of the result.**

***Index Terms*—Condition monitoring, change-point detection, operating regimes, fleet telemetry, haul trucks, statistical process control, novelty detection, honest benchmarking.**

I. Introduction

A HAUL truck loaded on an uphill ramp and the same truck empty on the way down are not the same machine as far as a detector is concerned. Every monitored channel moves with payload, grade and speed, and none of that movement is a fault. Raise the alarm threshold until the haul cycle stops firing and the fault you wanted sits under the threshold too. The remedy this report examines is the standard one from multivariate statistical process control: learn the operating regimes from *context* channels (payload, grade, speed), and monitor each telemetry channel as a within-regime residual,

$$r_t = \frac{x_t - \hat{\mu}_{k(t)}}{\hat{\sigma}_{k(t)}}, \tag{1}$$

with $k(t)$ the regime at time t, the statistics estimated only on the unit's own healthy baseline, and samples outside every seen regime left *unassigned* rather than snapped to the nearest one, because snapping is how a detector talks itself into calling a new operating condition a fault.

The idea is plausible enough to be sold without evidence, which is the problem. This report's contribution is not a new method; it is a controlled measurement of an old one, with the confounds removed, the nulls printed, one claim withdrawn, and one genuine counter-example reported at the same volume as the successes. The companion web application replays every number in this report from committed, versioned artifacts, and the detection engine is a separately published, MIT-licensed package (`regimecpd`) whose defect record is public and summarised in Appendix A.

II. Data, Protocol, Metrics

A. Four data lanes, with declared limits

No public, redistributable dataset of continuous haul-truck telemetry with labelled faults was found; that absence shapes the design and is a limitation, not a footnote. Four lanes stand in, each supporting a different part of the claim and none over-stretched. *(i)* NASA C-MAPSS turbofan data [1] supplies the controlled experiment: subsets FD001/FD003 run one operating condition, FD002/FD004 run six, with the condition declared per sample, so regime variation can be switched on and off while everything else is held fixed. *(ii)* The SCANIA APS dataset [2] supplies a decision problem under a published cost matrix (false positive 10, false negative 500). *(iii)* SCANIA Component X [3] supplies 23,550 real vehicles with a graded 5×5 cost matrix. *(iv)* A physically grounded synthetic haul fleet (resistance-driven cycle, strut pressure from payload, TKPH-style tyre heating; the regime confound *emerges* from the cycle rather than being injected) supplies the only lane where the true onset instant exists by construction. Nothing here is calibrated against a real truck, and the report never claims otherwise.

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Technical report, 2026. Interactive study and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_TruckVitals; live at https://truckvitals.fasl-work.com. The detection engine is the separate PyPI package `regimecpd` (https://github.com/fsantibanezleal/CAOS_RegimeCPD).

B. The five protocol rules

(1) The raw arm and the residual arm reach every detector as the same kind of object, so the detector cannot know which it is scoring. (2) Arms are compared at a matched *false-alarm budget*, never at a matched threshold: a z-scored residual and a raw channel in bar live on different scales. (3) Alarms are *events* (rising edges of the thresholded statistic), not samples: a statistic that sits above threshold for an hour is one alarm an operator answers, not sixty. The event-counted rate is not monotone in the threshold, and the threshold search must descend from the never-fires end; the first implementation ascended and selected a permanently-alarming detector whose single rising edge looked like a perfect rate. (4) Uncertainty is bootstrapped over *units*, never over samples, which would manufacture significance by raising the sample rate. (5) Nothing is fitted on the record it judges: regime model and residual statistics fit on a healthy baseline window, the threshold is chosen on a disjoint calibration window as a *fleet* quantity, and detection is scored on the remainder.

C. Metrics

Detection rate counts faulty units whose first rising edge is at or after the onset; an earlier alarm is a false alarm, not an early detection. Delay is the median over detected units only, and is always printed beside the rate. Onset localisation error is the distance to the *nearest* retrospective changepoint, which is optimistic by construction, so it is always divided by the chance level for the same changepoint count, $S/(2(k{+}1))$ for k cuts over a span S: skill 1.0 means no better than scattering the same number of cuts at random.

III. The Mechanism, With No Detector In It

Before any detector enters, the mechanism itself can be measured as an effect size that no threshold rule, alarm convention or budget can move. For each C-MAPSS subset, take the median over channels of the fault signature measured in pooled standard deviations, and the same signature measured in within-regime standard deviations, on the healthy fleet baseline; the per-unit fault shift cancels in the ratio, so the ratio is a property of the operating-regime structure alone.

Fig. 1 shows the result. On the multi-condition subsets the same physical fault is worth 0.16 pooled sigma, invisible, and 11.95 within-regime sigma, unmissable: ratios 90.2 (FD002) and 62.3 (FD004). The single-condition subsets return a ratio of exactly 1.00: with one regime, pooled and within-regime spreads are the same number, and the measurement says so to the last digit. That is a negative control nobody designed, and it is the reason to trust the instrument.

IV. Detection Under Regime Variation

Table I gives the controlled contrast: the same CUSUM detector (k=0.5), the same healthy data per arm, thresholds cross-fitted over units, and only the informative channels common to both subsets of a pair. Two confounds were found and removed before any number was believed: FD002 carries more informative channels than FD001 (and the statistic is a max across channels, so more channels raise the false-alarm rate on their own), hence the common-channel restriction; and fraction-based fit windows dropped units systematically differently per subset (38 versus 30 of 100), hence absolute cycle counts.

Three readings. First, regime variation alone destroys detection (0.93 → 0.17; 0.73 → 0.24), and conditioning recovers it to *above* the single-condition references (0.95; 0.90), at a lower realised false-alarm rate than the raw arm. Second, *discovering* the regimes by k-means on the context channels recovers exactly as much as being handed the declared condition labels (0.954 equals 0.954 to three decimals): the labels carry no information the context channels do not. Third, a global context regression with no segmentation at all recovers most of the gap (0.89; 0.88); on FD004 the margin between conditioning and plain regression is 0.02, which a careless report would round into a victory. These recovery figures moved from an earlier 0.98/0.70 when an engine review found that a NaN gap inside a sustained excursion was counted as a second alarm, a bias that fell on the residual arm by design;

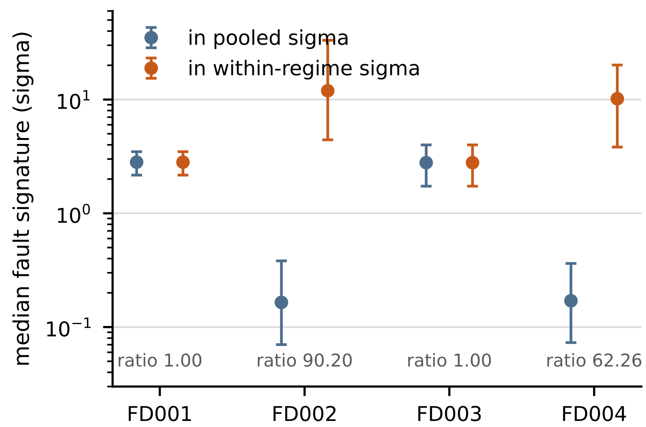

Fig. 1. The detector-free effect size. Median (p10–p90 over channels) fault signature per C-MAPSS subset, measured in pooled sigma (blue) and within-regime sigma (orange). Single-condition subsets (FD001, FD003) return a ratio of exactly 1.00, the built-in negative control; six-condition subsets return 90.2 and 62.3.

TABLE I
C-MAPSS detection at a matched false-alarm budget. Arms conditioned on a regime are bold. *global-regr* is a single global context regression (no segmentation), reported separately because an earlier version of this study wrongly credited its result to "regime conditioning".

Pair	Arm	Detection	Regimes
FD001 vs FD002	raw, 1 condition (ref.)	0.93	1
	raw, 6 conditions	0.17	–
	residual, observed	**0.95**	6
	residual, clustered	**0.95**	6
	global-regr, no segmentation	0.89	1
FD003 vs FD004	raw, 1 condition (ref.)	0.73	1
	raw, 6 conditions	0.24	–
	residual, observed	**0.90**	6
	residual, clustered	**0.90**	6
	global-regr, no segmentation	0.88	1

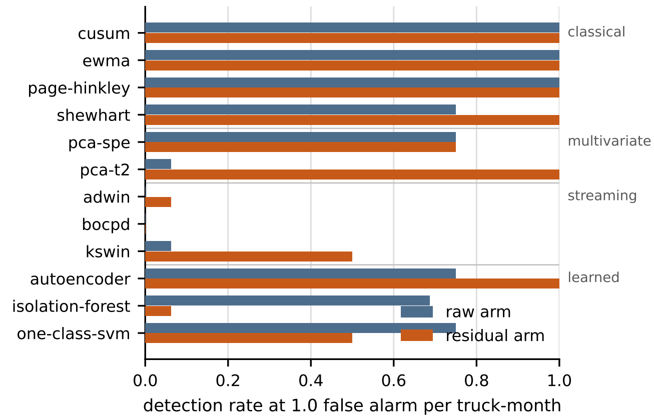

Fig. 2. The complete ladder at a matched budget, by tier. Conditioning helps five rungs (largest: Hotelling T^2, $0.06 \to 1.00$), leaves five at ceiling, and hurts two, both boundary-shaped learned novelty models. BOCPD detects nothing on either arm (gradual ramps against an abrupt-change model); ADWIN's integer statistic leaves it only $d{+}1$ operating points.

the artifacts were re-baked the same day and the earlier figures are not to be quoted.

V. The Ladder, and a Counter-Example

On the synthetic fleet (36 units, 16 faulty, four fault families injected with known onset), twelve online detectors run on both arms at the same budget of 1.0 false alarm per truck-month: Shewhart [4], CUSUM [5], EWMA [6], Page-Hinkley [5], [7] (reported as a CUSUM variant, not an independent baseline; at equal allowance the two correlate above 0.98), Hotelling T^2 and PCA SPE [8], [9], [10], BOCPD [12], KSWIN [14], ADWIN [13], isolation forest [17], one-class SVM [18] and a dense autoencoder [19]. PELT [15] and the multidimensional matrix profile [16] are deliberately *excluded*: both are retrospective (the profile value at a window is computed against matches anywhere in the record, including after it), and an online delay metric must never score a method that has read the future. A deliberately generous trivial baseline, a fixed threshold on the single channel each fault moves first, *told the answer*, scores 0.75 at 123 minutes; a ladder that cannot beat it would mean the lane is too easy to be informative.

Fig. 2 and Fig. 3 carry the result. The largest effect on the ladder is the mechanism made visible: raw T^2 detects 1 of 16 faulty trucks, because the retained principal subspace of a haul truck is dominated by the duty cycle, and a fault that moves the machine within that structure looks like a load change; the same statistic on residuals detects 16 of 16 at a $3.6\times$ shorter delay. SPE sits at 0.75 on both arms, and its gain is delay ($210.5 \to 86$ minutes): the quarter of faults it misses never breaks the correlation structure at this severity.

The counter-example is the finding this report exists to publish. Conditioning *hurts* the isolation forest ($0.69 \to 0.06$) and the one-class SVM ($0.75 \to 0.50$ on detection rate; its shorter conditioned delay conditions on the half it still catches). The obvious explanation, unassigned regime samples destroying the ten-sample feature windows these models score,

TABLE II
Onset localisation via nearest retrospective changepoint (36 units; medians). Skill is chance/error; 1.0 means no better than random cut placement.

Arm	Error (min)	Cuts	Chance (min)	Skill
raw	121.5	2	236.2	**1.94**$\times$
residual	52.5	13	50.6	0.96$\times$

is ruled out by measurement: regime coverage on the residual arm is 0.998, so there are almost no gaps to blame. The surviving explanation, offered explicitly as a hypothesis and not a result: a novelty model learns the *shape* of normal. On the raw arm that shape is a few tight regime clusters, and anything off-cluster is easy to isolate; conditioning replaces them with one roughly isotropic cloud, and a moderate shift stays inside it. The structure the residual removes is the structure these detectors were using. The autoencoder escapes on the same reading: its statistic is reconstruction error through a rank-limited map, so a fault that breaks cross-channel structure produces error regardless of how isotropic the healthy cloud looks, and it is the only learned rung conditioning helps ($0.75 \to 1.00$, $174 \to 90$ minutes) and the only one that beats the trivial baseline outright.

VI. Localisation Is a Null

Retrospective PELT segmentation estimates the onset instant on both arms; Table II shows why the chance column is the number to read. The conditioned arm has the smaller absolute error on every seed, by factors of two to seven, and all of that advantage is explained by producing 12–14 changepoints against 2–3: at thirteen cuts the chance level is already 50.6 minutes. Chance-corrected, the paired skill difference over five seeds is -0.08 ± 0.74, with conditioning ahead in two of five, and the ordering is not stable to fleet size (a 20-unit run puts conditioning at $2.40\times$ and the raw arm at $1.08\times$; the 36-unit run reverses it). An earlier version of this study shipped the 20-unit reading as a win; the chance baseline is what caught it. The claim ships as a null: regime conditioning does not reliably improve onset *localisation*, and this does not touch the detection result, which is a different measurement on different data, stable across detectors and subset pairs.

A withdrawal belongs beside the null: no false-alarm-reduction claim is made anywhere in this study. C-MAPSS measures detection at a *fixed* false-alarm budget, and on the synthetic lane the raw arm reaches the better false-alarm rate outright.

VII. Two Findings Not About Regimes

The cost lanes produced two results worth reporting even though no regime enters either. On SCANIA APS, with the dataset's own primary-verified cost matrix (FP 10, FN 500) and thresholds chosen out-of-fold, the F1-optimal decision rule attains the best F1 (0.807) and a total cost of 47,890, while the cost-optimal rule attains 11,670 at a worse F1 (0.652): optimising the metric almost everyone reports selects a decision $4.1\times$ more expensive than optimising the one the operator

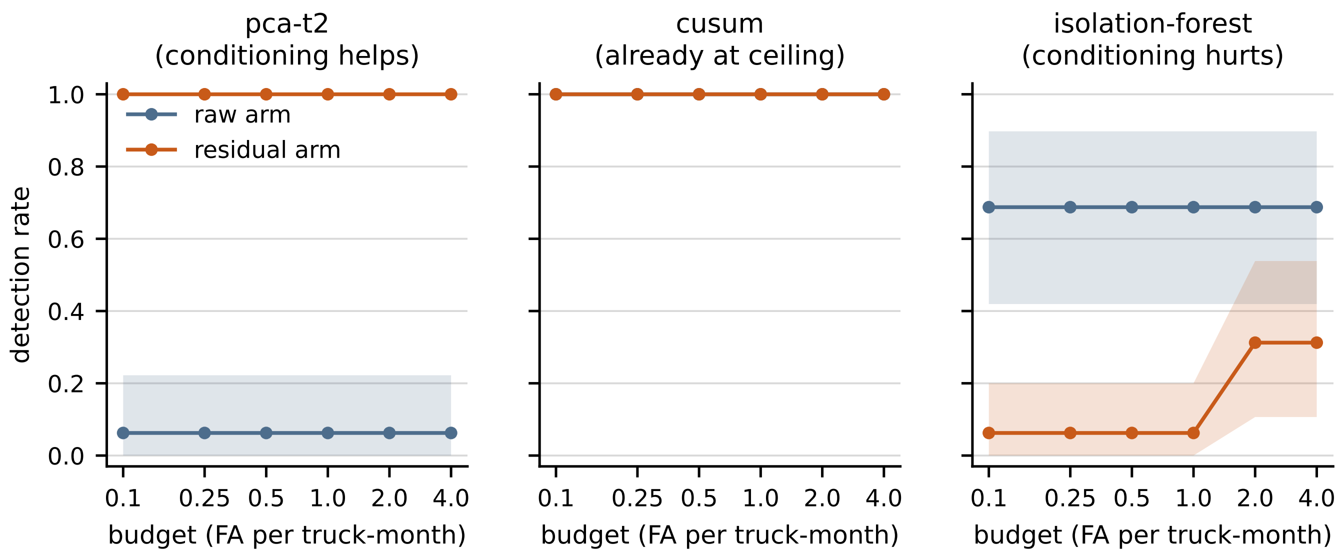

Fig. 3. Baked alarm-budget curves with bootstrap-over-units intervals for a rung conditioning helps (Hotelling T^2), a rung already at ceiling (CUSUM), and a rung conditioning hurts (isolation forest). The single-point ladder of Fig. 2 is one slice of these curves; a method that wins only at one budget has not won.

pays. On Component X, minimising expected cost under the graded matrix helps an unweighted model and degenerates to flagging every vehicle on a class-weighted one: class weights and a cost matrix encode the same asymmetry through two mechanisms, and applying both double-counts it, because an expected-cost decision needs calibrated probabilities and a class-weighted model's are not. Never flagging anything scores 97.3% accuracy and the worst cost of all, which is the same lesson from the other side.

VIII. Related Work

The components are deliberately standard: control charts and their multivariate extensions [4], [5], [6], [8], [9], [10], [11], Bayesian and windowed online change detection [12], [13], [14], retrospective segmentation [15], matrix profiles [16], healthy-only novelty models [17], [18], [19], and conformal calibration as the engine-level route to budget-comparable scores [20], [21] (implemented in the engine; the published numbers here run through direct event-counted threshold search instead, and the report says so rather than borrowing the guarantee). Closest in intent, Carpentier et al. [22] model heavy-duty trucks "contextually", but their context is a per-vehicle *cohort*: the fleet is clustered by specification and one model is trained per cohort, assigned once per vehicle. There is no per-timestep regime label and no residual channel. Their approach partitions the fleet; this one partitions the timeline. The two compose without interacting, and the full text was read before this paragraph was written.

IX. Limitations

The truck-specific results are synthetic: magnitudes are physically plausible for a large rigid hauler and are not measurements, and the missing real lane is the study's largest limitation. The controlled detection experiment lives on turbofans, chosen because the regime structure is declared and switchable, not because turbofans resemble trucks. The ladder runs at one severity and one seed (the seed sensitivity of the localisation null is exactly why it is a null); the budget sweep covers 0.1–4.0 alarms per truck-month. ADWIN's integer statistic makes it structurally ill-suited to a common budget, and its rows understate what a confidence sweep might show. The learned-tier counter-example is one lane and one feature contract; its explanation remains a hypothesis until it is tested on a second regime structure.

X. Conclusion

On load-varying telemetry, regime variation alone can destroy budgeted detection, and conditioning on regimes learned from context channels recovers it to above single-condition references, with clustering as good as declared labels. The same conditioning does not reliably improve onset localisation, earns no false-alarm claim, and actively hurts boundary-shaped novelty detectors while helping position and accumulation detectors. A practitioner reading this report should condition their control charts and T^2, keep their novelty detectors on the raw arm or re-validate them after conditioning, and distrust any localisation figure not printed beside its chance level.

Appendix A
The Defect Record

Every defect below produced a plausible number, not a crash, against a green test suite, and each is released and regression-tested in the public engine or product history. They are part of the result: a benchmark's honesty budget is set by the defects it went looking for.

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TABLE III
DEFECTS FOUND WHILE BUILDING THIS STUDY (RELEASE WHERE FIXED; MEASURED EFFECT).

Defect	Fixed; effect
A channel constant to within float noise (σ=7e−15) standardised by that noise; one unit set the fleet threshold	0.09.001; detection $0.79 \rightarrow 0.07$ before fix
ADWIN cut threshold matched neither the paper nor MOA while the docs said it did	0.09.002; one sample of median delay
Threshold search stopped at the first budget violation, blind to qualifying thresholds below	0.09.003; 14 thresholds unreachable, 0.046 vs 0.276 available
Degenerate-scale hole reopened in the linear residual branch (intercept forces the mean to zero)	0.09.004; dead channel drove 552/1199 samples
A NaN gap inside one sustained excursion counted as a second alarm, biasing the residual arm	0.09.005; recovery figures $0.98/0.70 \rightarrow 0.95/0.90$
Novelty scale floor referenced a signed mean, zero by construction on residuals	0.09.006; third module with the same defect class
ADWIN docstring still showed the pre-0.09.002 bound (found by equation-vs-code extraction)	0.09.007; docs-only
Raw arm given a third of the residual arm's healthy data	product; FD002 raw $0.046 \rightarrow 0.161$
Threshold chosen on the data it scored, while an unused `calibration` argument existed	product; no change in conclusion, real leak
Artifacts written with bare `NaN`, which is not JSON	product; the site could not parse its own headline artifact

[2] "IDA 2016 Industrial Challenge: APS failure at Scania trucks," UCI Machine Learning Repository, 2016, doi:10.24432/C51S51. The cost matrix (FP = 10, FN = 500) is primary-verified in the dataset's own description file.
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